

ARUL RHIK MAZUMDER

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Education

Carnegie Mellon University

Aug. 2023 – May 2026

Bachelor of Science in Computer Science, Dean's List High Honors

Pittsburgh, Pennsylvania

Select Coursework: Probabilistic Graphical Models, Quantum Computing, Complexity Theory,
Parallel Architecture and Algorithm Design, Deep Learning Systems

Activities: President of [CMU Quantum](#), Quant Club, ACM @ CMU, Project Olympus

Experience

Caltech SURF

May 2025 - August 2025

Undergraduate Research Fellow

Pasadena, California

- Worked under Samson Wang with John Preskill's research group to design early fault-tolerant quantum primitives based on Trotterization achieving exponential improvement in error on circuit depth compare to other modern techniques ([pdf](#)).
- Joined the quantum algorithms reading group, exploring ideas like the QSVT paradigm, Quantum Streaming, etc.

Duke Software-Tailored Architectures for Quantum Codesign (STAQ)

June 2025

NSF-Funded Invited Scholar

Durham, North Carolina

- Completed a PhD-level, week-long quantum computing seminar featuring advanced lectures on quantum simulation, error correction, and architecture, taught by leading researchers including Peter Love, Kenneth Brown, and Yu Tong.
- Participated in a lab tour at Duke Quantum Center and learned about experimental research across diverse quantum hardware platforms, including ion traps and neutral atoms.

International Business Machines (IBM)

May 2024 – August 2024

Research Intern

Cambridge, Massachusetts

- Researched a novel learning algorithm for discrete graphical models using samples generated from Glauber Dynamics.
- Implemented end-to-end sampling, learning, and evaluation pipelines in a dedicated research codebase ([GitHub](#)).

Carnegie Mellon University Quantum Technologies Group

May 2023 – Current

Research Intern

Pittsburgh, Pennsylvania

- Produced ([tutorial](#)) for quantum solutions to optimization problems on IBM and D-Wave machines ([GitHub](#)).
- To be ([presented](#)) at the 2025 INFORMs Annual Meeting and published in 2025 TutORials in Operations Research.

Select Projects/Publications

Quantum Annealing for solving the Shipment Rerouting | *Python, D-Wave Ocean, MILP*

January 2025

- Developed both classical approximation algorithms and quantum annealing algorithms for the shipment rerouting problem using D-Wave's quantum annealer and custom mixed-integer linear programming models ([paper](#)).
- Demonstrated quantum solutions achieving near-optimal performance faster than classical solvers in various scenarios.

Benchmarking Metaheuristic-Integrated QAOA against Quantum Annealing | *Python, Deep Learning*

May 2024

- Numerically benchmarked the performance of metaheuristic optimizers for optimizing QAOA circuits ([paper](#)).
- Published in Intelligent Computing Proceedings of the 2024 Computing Conference, Volume 3.
- Accepted to Quantum Information Processing (QIP) 2024 as Poster Presentation.

A Hybrid Quantum Algorithm for the Metric Traveling Salesman Problem | *Python, Ocean*

January 2023

- Researched and developed a novel quantum-classical hybrid algorithm for the Traveling Salesman Problem. Numerical tests on D-Wave backends for small input sizes demonstrate $3\times$ improvement ([paper](#)).
- Published in 2023 IEEE International Parallel and Distributed Processing Symposium (IPDPS).

Comparisons of Classic and Quantum String Matching Algorithms | *Python, Qiskit*

August 2022

- Developed novel quantum string-matching algorithm using Qiskit and tested it on IBM Quantum Backends ([paper](#)).
- Published in proceedings of ACM's 4th International Conference on Advanced Information Science and System (AISS).

Awards

Olympiads: USA(J)MO (2021-22), USAPhO Bronze (2022), USACO Gold (2023), USNCO National Finalist (2022)

Other Competitions: 2nd Place overall at YQuantum - The Yale Quantum Institute Grand Prize Recipient (2025),
2nd Place at Correlation One and Citadel Summer Invitational Datathon (2024),
Graduate Student Association Best Campus Infrastructure Hack Award at Hack@CMU (2023),
Regeneron Science Talent Search Scholar [Top 300] (2023), Purple Comet 5th Place (2022),
International Math Modelling Competition National Finalist (2022),
United States Math Talent Search Honorable Mention and Bronze (2021-22)